Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/21**

Paper 2 Theory **October/November 2020**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **11** printed pages.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

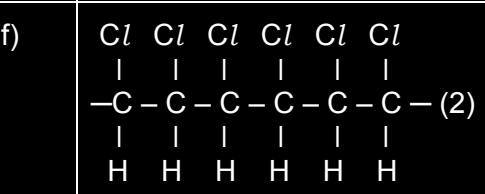
7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)	silver nitrate	**1**
1(b)	silver nitrate	**1**
1(c)	magnesium carbonate	**1**
1(d)	lithium bromide	**1**
1(e)	methane	**1**

Question	Answer	Marks
2(a)	carbon dioxide is a (simple) molecule **and** calcium carbide is ionic (1) for carbon dioxide: weak (attractive) forces between molecules / weak intermolecular forces(1) for calcium carbide: strong (attractive) forces between ions (1)	**3**
2(b)	$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$	**1**
2(c)	compound which contains carbon and hydrogen only / compound which contains carbon and hydrogen and no other element	**1**
2(d)	C_6H_{10}	**1**
2(e)(i)	addition / reduction / hydrogenation	**1**
2(e)(ii)	C_2H_4 / C_2H_6	**1**
2(f)	 if 2 marks not scored: 1 mark for this structure without extension bonds	**2**

Question	Answer	Marks
3(a)	(some of the) electrons can move / (some of the) electrons are mobile	**1**
3(b)	(aqueous) sodium hydroxide (1) (light) blue precipitate (insoluble in excess) (1) **OR** (aqueous) ammonia (1) (light) blue precipitate soluble in excess / dark blue solution formed in excess ammonia (1)	**2**
3(c)(i)	positive electrode: bubbles / effervescence / fizzes (1) negative electrode: turns pink (1) electrolyte: blue colour (of solution) fades (1)	**3**
3(c)(ii)	inert / (relatively) unreactive	**1**
3(d)	$Cu^{2+}(aq) + Mg(s) \rightarrow Cu(s) + Mg^{2+}(aq)$ correct formulae and balancing (1) correct state symbols – dependent on correct formulae (1)	**2**
3(e)	mass of copper = 2.00 g (1) mol copper = $\frac{2.00}{64}$ **and** mol oxygen = $\frac{0.25}{16}$ **OR** mol copper = 0.0313 **and** mol oxygen = 0.0156 (1) empirical formula is Cu_2O (1)	**3**

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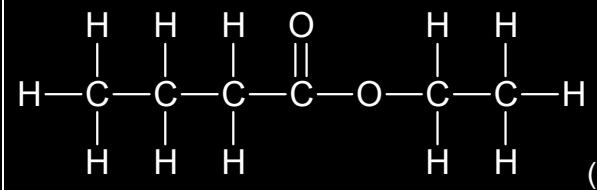
Question	Answer	Marks
3(f)	mixture of metals / mixture of a metal and another element	**1**

Question	Answer	Marks
4(a)	any **three** from: • evaporation • particles with highest kinetic energy escape the liquid / particle moving the fastest escape the liquid • diffusion • particles move (from place to place) / particles collide • random (movement) of particles / particles go anywhere / particles (move) in all directions / particles disperse • intermingling of particles / mixing of particles • (bulk movement of bromine particles) from higher to lower concentration	**3**
4(b)(i)	darker / lighter up the Group	**1**
4(b)(ii)	chlorine gas **and** iodine solid	**1**
4(b)(iii)	disinfectant / killing bacteria / bleach	**1**
4(c)	I_2Cl_6	**1**
4(d)	mol oxygen = $\frac{0.037}{4}$ **OR** 9.25×10^{-3} (1) volume of oxygen = 0.22 (dm^3) (1)	**2**

Question	Answer	Marks
5(a)	any **two** from: • high density (1) • form coloured compounds (1) • have variable oxidation states (1) • have catalytic activity (1)	**2**

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Question	Answer	Marks
5(b)(i)	lowers the activation energy	**1**
5(b)(ii)	high temperature (1) low pressure (1)	**2**
5(c)	$3V_2O_5$ + **10**Al → **6**V + **5**Al_2O_3	**1**
5(d)(i)	fractional distillation	**1**
5(d)(ii)	<u>reduction</u> because electrons gained / <u>reduction</u> because electrons added	**1**

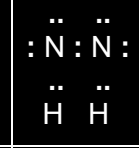
Question	Answer	Marks
6(a)(i)	(density) increases	**1**
6(a)(ii)	(boiling points) increase because forces of attraction between molecules are stronger / (boiling points) increase because the intermolecular forces increase	**1**
6(b)(i)	(acid which is) completely ionised (in water) / (acid which is) completely dissociated (in water)	**1**
6(b)(ii)	ethyl butanoate (1)  (1)	**2**
6(c)	(acidified) potassium manganate(VII) (1) reflux / heat (1)	**2**
6(d)(i)	$2CH_3CO_2H$ + Ca → $(CH_3CO_2)_2Ca$ + H_2	**1**

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Question	Answer	Marks
6(d)(ii)	rate decreases because: fewer particles in a given volume / particles further apart / fewer particles per unit volume (1) fewer frequent collisions / particles collide less often / collision rate increases (1)	**2**

Question	Answer	Marks
7(a)(i)	aluminium is (very) high in the reactivity series / aluminium ions accept electrons less readily than carbon	**1**
7(a)(ii)	to dissolve the aluminium oxide	**1**
7(a)(iii)	$2O^{2-} \rightarrow O_2 + 4e^-$	**1**
7(b)(i)	oxygen removed from aluminium oxide / it loses oxygen	**1**
7(b)(ii)	mol aluminium oxide = $\frac{25.5}{102}$ **OR** 0.25 mol (1) mass of aluminium = 13.5 g (1)	**2**
7(c)	(positive) ions in layers (1) layers (easily) slide over each other (1)	**2**
7(d)	bond breaking endothermic and bond making exothermic / heat absorbed to break bonds and heat released on making bonds (1) more heat released than absorbed / more energy released than absorbed (1)	**2**

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Question	Answer	Marks
8(a)(i)	mol barium hydroxide = $0.045 \times \frac{34.0}{1000}$ **OR** 1.53×10^{-3} (1) mol nitric acid = 3.06×10^{-3} (1) concentration nitric acid = 0.122(4) mol / dm^3 (1)	**3**
8(a)(ii)	evaporate solution until crystallisation point / evaporate until solution is saturated (1) filter off crystals **AND** wash with organic solvent / wash with cold water (1) dry crystals with filter paper (1)	**3**
8(b)	$2Ba(NO_3)_2 \rightarrow 2BaO + 4NO_2 + O_2$	**1**
8(c)	erodes (buildings) / corrodes (metalwork) / reacts with (mortar)	**1**
8(d)(i)		**1**
8(d)(ii)	0°C is below the melting point / the melting point is above 0°C	**1**

Question	Answer	Marks
9(a)	Add (acidified) potassium manganate(VII) (1) goes from purple to colourless (1)	**2**

Question	Answer	Marks
9(b)	electrons:18 (1) neutrons: 24 (1)	**2**
9(c)(i)	equilibrium moves to the right / more products formed (1) concentration of carbon dioxide falls so more calcium carbonate decomposes to (try to) maintain equilibrium (1)	**2**
9(c)(ii)	equilibrium moves to the right / more products formed (1) increasing temperature pushes the reaction in the direction of absorbing energy / increasing temperature pushes the reaction in the direction of the endothermic reaction (1)	**2**
9(d)	$2CaO + \mathbf{2}Cl_2 \rightarrow \mathbf{2}CaCl_2 + \mathbf{O_2}$	**1**
9(e)	any suitable soluble calcium salt other than calcium chloride e.g. calcium nitrate, calcium ethanoate	**1**

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